

Grammaticality as Kind: Ontology, Epistemology, and Empirical Pay-off

Brett Reynolds ¹ and J. T. M. Miller ²

¹Humber Polytechnic & University of Toronto, brett.reynolds@utoronto.ca

²Department of Philosophy, Durham University, james.miller@durham.ac.uk

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Abstract

This paper argues that grammaticality is best understood as a homeostatic property-cluster (HPC) kind anchored in morphosyntactic form–meaning pairings and regimented by community practice. The framework distinguishes objective grammatical status – whether the community licenses a form–meaning pairing – from the subjective feeling of ungrammaticality, which integrates grammaticality with processing costs, social evaluation, and prescriptive norms. Drawing on contemporary metaphysics of science, we show how grammaticality exhibits the key features of HPC kinds: a cluster of co-occurring properties (morphosyntactic well-formedness, semantic compatibility, community entrenchment), homeostatic mechanisms that maintain cluster stability (language acquisition, repair strategies, sociolinguistic policing), and systematic projectability supporting inductive generalization. Processing constraints shape what patterns can be licensed through acquisition and transmission while also affecting how licensed patterns feel during comprehension. This analysis clarifies why grammaticality supports explanatory and inductive generalization despite surface variation and gradience. The HPC framework predicts which phenomena will pattern with grammaticality (morphosyntactic violations, semantic mismatches at the syntax-semantics interface) versus those that won't (pure phonotactic violations, processing overload). By grounding grammaticality in both cognitive mechanisms and social practices, the account reconciles the apparent universality of particular constraints with the evident community-relativity of others. The framework has implications for methodology (validating judgment tasks while acknowledging their limits), theory (reconciling generative and usage-based evidence), and typology (predicting which “exotic” grammars can exist as stable linguistic systems).

Keywords: grammaticality; homeostatic property clusters; linguistic ontology; acceptability judgments; morphosyntax; philosophy of linguistics; typology

1 INTRODUCTION

Consider three utterances that native English speakers consistently reject:

- (1) a. * *I've finished it yesterday.*
- b. * *How many years do you have?* [intended 'how old are you']
- c. * *Furiously sleep ideas green colorless.*

Each violates something we call “grammaticality”, but the violations differ profoundly. The first shows a semantic mismatch between tense (present) and past-time adverbial, the second seems semantically coherent but fails to match community expectations, and the third fails to evoke any coherent meaning. Meanwhile, speakers accept the semantically bizarre *Colorless green ideas sleep furiously* and trained linguists classify the unprocessable *The rat the cat the dog chased killed ate the cheese* as “grammatical”.

This variation poses a fundamental question: what kind of thing is grammaticality? The question isn't merely terminological. How we answer determines what counts as evidence, which methods are appropriate, and what explanations succeed. If grammaticality counts as a genuine kind in Khalidi's (2013) causal-realist sense, then its explanatory machinery will differ radically from a purely social construction. How we conceive of grammaticality shapes the evidence we look for and the explanations we accept.

This paper argues that grammaticality is best understood as a homeostatic property-cluster (HPC) kind. Following R. Boyd (1991), HPC kinds are individuated not by necessary and sufficient conditions, but by the regular clustering of properties instantiated by members of the kind, where those properties regularly cluster due to the acting of homeostatic mechanisms. It is the mechanism-sustained regularity of clustering that makes HPC kinds projectable – and thus epistemically significant (Khalidi, 2013).

For grammaticality, the property cluster includes morphosyntactic well-formedness, semantic compatibility, and community entrenchment. The homeostatic mechanisms include child language acquisition, adult repair strategies, and sociolinguistic policing. Together, these create sufficient stability for grammaticality to function as a scientific kind despite exhibiting gradience, variation, and change. Processing constraints shape what patterns these mechanisms can maintain while also affecting how licensed patterns feel during comprehension (§5.1).

The framework provides a principled basis for distinguishing grammatical status (what the community licenses) from acceptability judgments (how constructions feel during processing). This explains why processing-heavy strings can be grammatical yet feel unacceptable, and why frequent errors can feel acceptable yet remain unlicensed.

We also argue that the HPC analysis can suggest a way to resolve some persistent puzzles, validates methodological pluralism concerning grammaticality, and supports predictions in typology.

2 THE EMPIRICAL LANDSCAPE: WHAT ANY ONTOLOGICAL ACCOUNT MUST EXPLAIN

Linguists appeal to grammaticality as if it were a stable ontological target, but persistent disagreements about its nature suggest the concept needs clearer metaphysical grounding. Is grammaticality a matter of rule compliance, usage frequency, cognitive competence, or something else entirely? The answer determines what counts as evidence in linguistic theory, which explanations succeed, and how we understand the relationship between individual knowledge and community norms. This section

surveys the empirical profile of grammaticality: a cluster of properties that co-occur systematically, remain stable across measurement contexts, but dissociate in predictable ways. The cluster supports robust inductive generalization despite surface variation, and it resists reduction to essentialist definitions à la Kripke (1980) and Putnam (1975), pure social construction, or cognitive mechanism alone.

At the foundation, speakers treat systematic pairings between morphosyntactic forms and meanings – from morphemes to clause-level argument-structure templates – as part of the shared system of form–meaning resources. The English present perfect construction carries a specific temporal-aspectual meaning (current relevance); argument structure frames come with predictable role assignments; agreement morphology realizes feature bundles. These pairings aren’t arbitrary associations that vary freely across speakers. They survive lexical change and style-shifting, generalize to new items, and show up consistently in acquisition, repair behavior, and cross-dialectal comparison (Goldberg, 1995; Heim & Kratzer, 1998).

Grammaticality judgments track clashes between what the morphosyntactic form *means* and what the utterance as a whole requires. Consider the temporal incompatibility in **I’ve finished it yesterday*. The present perfect’s current-relevance meaning jars with the adverb’s completed-past specification. This isn’t lexical-semantic anomaly – *Colorless green ideas sleep furiously* is conceptually bizarre but grammatical. The licensing work is done by compatibility between morphosyntactic meaning and compositional interpretation, not by string well-formedness alone (Klein, 1992; Kratzer, 1998).

Argument-structure coercion shows the pattern more generally. English permits *She kicked him the ball* (caused-motion reading), but **She disappeared him the evidence* fails because the ditransitive frame requires the verb to denote a caused-transfer or caused-possession event, and *disappear* resists that coercion. The surface frame can be impeccable while the pairing is still rejected. We can operationalize this dimension via systematic judgment elicitation, repair patterns, and cross-linguistic comparison of where form–meaning alignments succeed versus fail.

Acceptance depends on conventionalization within a speech community, not raw exposure counts. This dissociates grammaticality from mere frequency: high-exposure errors like **The keys to the cabinet is here* don’t entrench despite occurring often in performance (Wagers et al., 2009), while some low-frequency patterns do entrench when they instantiate licensed pairings (Bybee, 2006). Multiple independent measures track this dimension: production choices and self-repairs, corpus distributions, recognition memory, and social-evaluative responses (Schmid, 2020). Together they reveal how pairings acquire and maintain community status.

The independent relative *whose* provides a revealing case. While *whose* functions perfectly well in dependent relatives (*the student whose paper was late*) and as an interrogative (*Whose is that?*), its use as an independent relative pronoun (*?I saw Joan, a friend of whose was visiting*) remains marginal. The construction is so rare that many speakers never encounter it, preventing stable conventionalization even though no widely accepted formal prohibition is agreed upon. This pattern challenges pure frequency accounts – entrenchment can’t reduce to exposure (§6 returns to this point).

These three properties – pairings, compatibility, and entrenchment – form the licensing core. Processing constraints shape what patterns can be stably maintained: they filter patterns exceeding learners’ capacities and amplify learnability biases across generations. Real-time processing difficulty also affects how licensed patterns feel during comprehension. A licensed but processing-heavy structure can feel unacceptable; an unlicensed but easy string can slip past detection (§5.1). Multiple center-embeddings provide the canonical case: *The bread the baker the apprentice helped made is delicious* is

structurally licensed but consistently judged poorly (Gibson, 2000, §2.2.8). Repetition and practice can improve ratings for processing-heavy but licensed constructions – a change in subjective response, not licensing status.

This dissociation between grammatical status and acceptability appears consistently across experimental paradigms. Self-paced reading studies show that comprehenders can successfully parse structures they rate as unacceptable, while some highly acceptable strings turn out to violate formal constraints when analyzed carefully (Gibson, 1998; Schütze, 2016). Known cases of illusions – where objective grammaticality and subjective feeling dissociate – show method-specific dissociations without threatening the underlying property cluster (Fanselow, 2021; Phillips et al., 2011).

Some configurations resist licensing regardless of context, frequency, or repeated exposure. English left-branch extraction (**Which did you buy car?*) remains categorically rejected across tasks, speakers, and presentation modes, and cross-linguistic analogues appear in genetically unrelated languages. Unlike processing difficulties that diminish with familiarity or compatibility mismatches that context can resolve, these bans persist across presentation modes and familiarity manipulations. Satiation experiments that successfully improve ratings for other marginal constructions leave these violations untouched (Snyder, 2022). Whether these bans reflect autonomous syntactic constraints or emerge from the interaction of other factors remains contentious; the present analysis is agnostic about their source while recognizing their role in the property cluster (§5 returns to this).

Judgments can shift when forms index inappropriate stance, identity, or social setting. The *vos* forms standard in Río de la Plata Spanish may be judged ungrammatical by speakers from *tú*-only regions, not because the morphosyntax fails but because the situation doesn't license that pairing for those speakers. Indexical mismatch modulates judgments but doesn't by itself generate licensing failures absent morphosyntactic or compatibility conflict. This doesn't erase the special role of morphosyntactic form–meaning relations; it shows that grammaticality emerges where morphosyntactic meaning, composite interpretation, and situation-licensing converge (Eckert, 2012).

The foregoing doesn't mean form and meaning are inseparable. Reynolds (2025a) demonstrates that morphosyntactic form and semantic interpretation can constitute distinct kinds that overlap substantially while remaining causally independent. The analysis of English definiteness shows how structural categories (distributional properties of determiners) and semantic categories (referent identifiability) correlate strongly through grammaticalization and discourse efficiency yet systematically dissociate in weak definites, generic definites, and proper names. The form–meaning pairing that matters for grammaticality involves the morphosyntactic component of such pairings – the systematic mapping between structural patterns and compositional meanings – rather than requiring perfect alignment between form and interpretation.

These properties exhibit two further patterns demanding ontological explanation.

The first of these is cross-method convergence. Acceptability ratings correlate with production frequencies, reading times, and neural signatures. Sentences rated as ungrammatical show increased P600 ERP components – often associated with syntactic reanalysis or integration rather than a single mechanism (Kaan et al., 2000) – and speakers avoid producing them in elicited tasks (Osterhout & Holcomb, 1992). The methods disagree on boundary cases but converge on clear violations, and known cases of grammatical illusions show method-specific dissociations without threatening the cluster. This convergence suggests we're tracking a real phenomenon rather than measurement artifact.

The second is bounded cross-linguistic variation. Languages show strong convergence on certain core constraints while varying systematically in others. Left-branch extraction prohibitions recur across genetically unrelated languages, and typological surveys reveal that languages cluster in accessible regions of grammatical space rather than distributing uniformly across logical possibilities (Bickel, 2007; Hawkins, 2004; Newmeyer, 2005). Certain imaginable grammatical systems – those requiring arbitrary long-distance dependencies without locality constraints, or combining completely free word order with no case marking – are so far systematically absent in attested inventories. This suggests that grammatical systems can only vary along certain evolutionary pathways, constrained by learnability and processing architecture. We reserve “compatibility failure” for syntax–semantics interface mismatches that do bear on grammaticality (§2, §4) and “lexical-semantic anomaly” for conceptual oddity that does not (§5.3).

The preceding discussion indicates a distinction between two conceptually distinct levels: objective grammaticality and the subjective feeling of (un)grammaticality.

Whether a form–meaning pairing is licensed within a speech community is determined by the three core properties already noted: successful morphosyntactic pairing, compatibility between meanings, and community entrenchment. Grammaticality in this sense is the objective target: whether the community licenses a given form–meaning pairing.

This is distinct from the metacognitive response signalling that “something is wrong.” This feeling integrates grammatical status with additional factors: processing costs, social evaluation, prescriptive awareness, and contextual appropriateness. Acceptability ratings primarily tap this subjective dimension. Throughout we contrast grammaticality with adjacent constraint regimes – phonotactics (§5.2) and lexical semantics (§5.3) – whose signals are integrated into the subjective feeling of (un)grammaticality but are not constitutive of licensing.

Three patterns reveal the importance of separating these levels. The first two involve what Fanselow (2021) calls illusions of (un)grammaticality – systematic mismatches between licensing and feeling.

First are grammaticality illusions: centre-embedded structures like *The rat the cat the dog chased killed ate the cheese* show intact form–meaning pairings (each clause is structurally licensed and semantically interpretable) but trigger strong negative feelings due to processing overload. Speakers consistently reject these strings in acceptability tasks yet demonstrate knowledge of the underlying pattern when processing simpler versions. The grammatical system licenses the pairing; working memory limits prevent comfortable processing.

Second are ungrammaticality illusions: agreement attraction errors like **The key to the cabinets are here* occur frequently in performance and often pass undetected in comprehension (Wagers et al., 2009). The feeling system fails to flag the violation when interfering material (plural *cabinets*) misleads the parser. But the construction remains unlicensed – no amount of exposure converts it to grammatical status, and meta-linguistic reflection consistently identifies it as erroneous.

A third related point is that novel constructions can initially feel odd while remaining structurally licensed. With repeated exposure in appropriate contexts, the negative feeling diminishes – a change in subjective response. If the construction additionally becomes conventionalized within the community, it shifts from marginal to fully grammatical – a change in licensing status. Acceptability ratings track both processes but conflate them, making the distinction empirically crucial.

Acceptability judgment tasks measure the feeling dimension, not grammaticality directly. This explains several puzzling patterns in experimental syntax. Speakers provide intermediate scores for

constructions that are categorically grammatical or ungrammatical because additional factors – processing difficulty, register appropriateness, frequency – modulate the feeling while leaving licensing status intact (Francis, 2022; Sorace & Keller, 2005). This explains gradient ratings without abandoning categorical distinctions.

Repeated exposure can improve acceptability ratings without changing grammatical status (Lu et al., 2024; Snyder, 2000, 2022). The construction becomes easier to process (feeling improves) while remaining unlicensed by the community (grammaticality unchanged). But for constructions that are grammatical yet rare, satiation reveals their underlying licensing by familiarizing speakers with an already-licensed pattern. These satiation effects show the licensing/feeling distinction in action.

Two speakers might agree that a construction is unlicensed (both judge it ungrammatical upon reflection) while providing different acceptability ratings (one processes it with more difficulty than the other). This cross-speaker variation reflects individual differences in working memory, exposure history, or prescriptive training, not disagreement about community licensing.

This dual-level structure is empirically well supported. Processing costs, prescriptive norms, and social evaluation are real factors that influence judgments without determining grammatical status. Recognizing both levels prevents conflating competence with performance while validating acceptability judgments as useful (though indirect) evidence about grammaticality.

Three features of this empirical profile demand an ontological account rather than merely descriptive taxonomy. First, we find *stable clustering with partial independence*. The properties co-occur systematically yet remain dissociable. A licensed but processing-heavy string can feel unacceptable; an unlicensed but easy string can slip past detection.

Second, there is *causal maintenance across timescales*. The property cluster doesn't persist by accident. Identifiable mechanisms operating at different timescales – from millisecond-level processing to generational transmission – actively maintain the clustering. This causal efficacy distinguishes genuine kinds from arbitrary groupings (Khalidi, 2013).

Third, there is *systematic projectability*. The cluster licenses robust induction. We can predict that satiation will shift ratings for rarity-driven or processing-driven marginality but not for categorical bans. We can predict that high-exposure agreement errors will remain unlicensed. We can predict which novel constructions will spread when social conditions favor them. This predictive power is the epistemic hallmark of scientifically important kinds – they support explanations and counterfactuals that mere classificatory schemes don't (R. Boyd, 1991).

Neither of the two standard ontological accounts can accommodate this profile. Essentialism holds that kind-membership is determined by necessary and sufficient conditions, but no such conditions explain why morphosyntactic well-formedness, semantic compatibility, and community entrenchment co-occur yet dissociate in predictable ways (Croft, 2001; Tomasello, 2003). Conventionism holds that grammaticality is whatever a community agrees it to be, but pure convention cannot explain why high-frequency errors like *could of* for *could have* persistently fail to entrench (Han, 2004), why processing costs constrain grammatical patterns independently of social agreement, or why structural bans converge across unrelated languages (Sprouse et al., 2016). Each account captures part of the picture – essentialism explains the categorical core, conventionalism explains community variation – but neither can explain clustering, causal maintenance, and projectability together.

We will argue that a homeostatic property-cluster (HPC) view of kinds can. Section 3 reviews the HPC framework; §4 shows that grammaticality satisfies its three criteria.

3 KINDS IN CONTEMPORARY METAPHYSICS

The previous section identified grammaticality's distinctive profile: a cluster of properties that co-occur systematically, remain stable through causal mechanisms, and support inductive generalization. This profile resists reduction to essentialist definitions or purely social construction. But what ontological framework can accommodate clustering without essences, causal maintenance without reduction, and variation within systematic bounds?

The homeostatic property-cluster (HPC) view of kinds, developed by R. Boyd (1991) and R. N. Boyd (1990, 1999a, 1999b), provides exactly this. HPC kinds are individuated not by necessary and sufficient conditions but by clusters of properties maintained through identifiable causal mechanisms. The view is widely accepted for biological species (Ereshefsky & Reydon, 2015) and has been extended to linguistic phenomena such as word-kinds (Miller, 2021). Before turning to grammaticality (§4), we first spell out what makes something an HPC kind. We contrast this with essentialist approaches (Kripke, 1980; Putnam, 1975) and explain why the HPC framework outperforms them.

One traditional view of kinds, which can trace its roots back to Aristotle, is essentialism (Kripke, 1980; Putnam, 1975). An essentialist holds that natural kinds are real, with membership of a kind tracking genuine distinctions in the world, and with a kind being defined by its essence. This means that a kind is defined in terms of certain essential properties such that all instances of a kind share some property (or set of properties) in common, and that property (or set of properties) provides the necessary and sufficient conditions of being an instance of that kind. For example, we might hold that the kind GOLD is defined by the property of having an atomic number of 79, such that having that property is a necessary and sufficient condition on some particular entity being an instance of that kind.

It has already been argued that essentialism can't easily be applied to various cases – human kinds, biological kinds (such as species) and linguistic phenomena – and similar concerns arise when we try to understand grammaticality within an essentialist framework. Put simply, it's not clear what property (or properties) all grammatical (or ungrammatical) utterances share such that the instantiation of that property (or properties) can provide the necessary and sufficient conditions for grammaticality. As we have already outlined in this paper, attempts to use rule compliance, usage frequency and competence all fail due to the variability in how utterances can be deemed ungrammatical. While various proposals have been made in the literature, and all capture some set of ungrammatical utterances, none capture all and hence providing an essentialist account of (un)grammaticality is empirically implausible.

Our proposal in this paper is that while essentialism fails, an alternative metaphysics of kinds can allow us to progress in our understanding of grammaticality. The rest of this section outlines this alternative proposal, with following sections applying it more specifically to grammaticality.

The homeostatic property cluster (HPC) view of kinds was developed in a series of papers by R. Boyd (1991) and R. N. Boyd (1990, 1999a, 1999b), and has been widely accepted in various domains, especially within biology and the philosophy of biology as providing a good account of species (Ereshefsky & Reydon, 2015). The HPC view rejects the possibility of using necessary and sufficient conditions to define a kind, but aims to maintain the idea that members of a kind share certain clusters of properties in virtue of being a member of that kind, whilst being able to accommodate the variation that we see within kinds. What makes two particular entities members of the same kind is that they instantiate some subset of a cluster of properties, and that those properties are instantiated due to the

acting of certain homeostatic mechanisms. It's these homeostatic mechanisms that ensure that there are regular, projectible, and scientifically tractable clusters of properties in nature, despite variations in the properties instantiated by members of the same kind. This means that for the HPC view, a kind isn't defined solely by members instantiating some shared cluster of properties, but instead by the mechanisms that give rise to members of a kind instantiating those properties.

This allows the HPC to accommodate more flexibility. The relevant homeostatic mechanisms that partly define a kind need not act such that every member instantiates all the properties within the cluster of properties. Individual members will vary, but will remain members of the same kind if those members instantiate properties from the same cluster of properties due to the acting of the underlying homeostatic mechanisms that cause and sustain those properties to cluster.

Kinds, under the HPC view, play an important epistemic role within our theorizing about the world in virtue of the clustering of properties which explain the causal powers of kinds. Members of a kind have similar causal powers because they instantiate properties from the same cluster. This means that the HPC view is still taken to be a realist account of kinds. As Khalidi (2013) argues, an important aspect of real or genuine kinds is that they're associated with scientifically important properties. Kinds permit inductive inferences and generalizations that allow us to explain naturally occurring phenomena. Under the HPC view, such inferences and generalizations can be maintained due to the continued acting of the underlying homeostatic mechanisms, but kinds aren't eternal (contra Platonic versions of essentialism) nor static as it may be that, over time, either the underlying mechanisms change, resulting in new sets of properties clustering, or even cease to act entirely.

To illustrate, consider species-kinds. Most biologists (and philosophers of biology) now accept that there are no genotypic or phenotypic properties that are both necessary and sufficient for belonging to a species (Ereshefsky & Reydon, 2015). Still, it's clear that there is important clustering of properties within members of a species. For example, the Golden Mantella (*Mantella aurantiaca*) is a small frog species endemic to Madagascar. Members of this species have certain properties in common – they eat small invertebrates and are poisonous, for instance. The HPC view holds that these (and other) properties are properties that contingently cluster in those individuals, despite other variation between individuals of this species. For instance, some have yellow skin (giving rise to the English name of the species), others orange or red.

We therefore have a certain set of properties that regularly cluster within individuals of this kind, and, importantly, we can start to provide the developmental, evolutionary, and environmental mechanisms the acting of which gives rise to the regular clustering of those properties. No property, nor even the set of properties, is essential as there is no property (or set of properties) that all members of the kind share, but the kind GOLDEN MANTELLA is still a real kind, defined by the cluster of properties and the underlying homeostatic mechanisms, such that the kind underpins various inductive inferences and generalizations.

This framework has also been extended to words, which one account treats as HPC kinds. On that view, word-kinds can be “defined by a cluster of properties, and the mechanisms underlying those properties” (Miller, 2021, p. 20). For example, the word TREE is a kind, with various instances (*tree*, *tree*, & *tree*) as members of this kind. As before, these instances instantiate properties that cluster in regular, projectible ways: the instances above all have the orthographic property of being spelt ⟨tree⟩, and expressing the semantic content “tree”. Such properties can't, though, define the kind TREE as some members of the kind fail to have such properties. Rather, what defines the kind – what determ-

ines if some word-token is a member of the word-kind *TREE* – is that the token instantiates some of the relevant cluster of properties, and that those properties arise due to the acting of underlying homeostatic mechanisms.

As will be important later for our discussion of grammaticality, these mechanisms can be internal or external. Internal mechanisms are those that “are responsible for deriving sentences such that the lexical item ‘table’ is used in a regular and consistent way. Such mechanisms are internal to an individual and ensure that an individual can retrieve lexical items regularly and reliably from their mental lexicon” (Miller, 2021, p. 25), while external mechanisms are those that reflect the wider community of speakers that an individual is part of. Internal mechanisms therefore ensure that we, as individuals, create word-tokens that instantiate regular, projectible clusters of properties, while external mechanism ensure that the word-tokens instantiate clusters of properties that align with community performance standards.

In the rest of this paper, we apply the HPC view of kinds to grammaticality itself. We will argue that grammaticality, qua kind, can be defined through a stable constellation of morphosyntactic pairings, compatibility relations, processing biases, and community entrenchment, kept in place by the acting of certain homeostatic mechanisms, both internal and external to the individual. Through this, we argue that grammaticality can be analysed as a causally real kind, underpinning inductive inferences and generalizations in a way that maintains the stability of the kind while accommodating the observed variation.

We note that naturally this kind, like word-kinds also, are clearly social kinds. That is, linguistic kinds like these are reliant on the existence of humans and our patterns of behaviour, unlike, at least arguably, other naturally occurring kinds. But this does not exclude this kind from being classed as “natural”. As Mason and Ritchie write, “Social kinds, like the human beings who create them, are occupants of the natural world, and are subject to the same physical laws that govern the behavior of everything from planets to protons” (Mason & Ritchie, 2020, p. 318). Just like other kinds – including other linguistic kinds – grammaticality, we will argue, is a kind that can be studied empirically, allowing us to predict and explain empirical phenomena. Social entities aren’t excludable from the domain of metaphysics.

Indeed, grammaticality is a particularly interesting case. Grammaticality can’t be reduced to pure social construction in that we can’t simply decide what is grammatical and what isn’t. There are limits on what communities can conventionalize, indicating that there is at least some non-social element to grammaticality. But, as noted, grammaticality is clearly not entirely non-social. Grammaticality, as a kind, appears to bridge the social/non-social divide. The HPC framework is uniquely suited to capturing this hybridity: it accommodates both cognitive constraints (through acquisition and processing mechanisms) and social dynamics (through community entrenchment and prestige selection) without reducing either to the other.

4 WHY GRAMMATICALITY FITS THE HPC TEMPLATE

Having set out grammaticality’s empirical profile (§2) and the homeostatic property-cluster (HPC) framework (§3), we now argue that grammaticality fits the HPC template at the system level. Two claims need separating. Claim A: Many individual constructions (passives, ditransitives, extraction

templates) are HPC kinds sustained by the five mechanisms we identify.¹ Claim B: Grammaticality, as a property of the system, is an HPC kind. Our target is Claim B. It presupposes, but does not reduce to, Claim A.

The step from A to B is a part-whole observation.² System-level properties of grammaticality are not recoverable from any single construction description. Knowing everything about the passive, the ditransitive, and the relative clause individually does not tell us that the *same* locality constraints govern all three, or that combining extraction with intervening nominals produces penalties exceeding the sum of each factor. These are properties of the *system of constructions*, maintained by the same five mechanisms operating across the system rather than within any one template.

Three classes of evidence substantiate this system-level claim. First, *cross-template generalization*: constraints identified on one construction type (e.g., interrogatives) predict licensing on others (e.g., relatives, comparatives) when predictions track structural descriptors like locality rather than construction labels. Structure-sensitive priming demonstrates abstraction over syntactic structure independent of lexical overlap (Bock, 1986). Factorial island experiments show that island constraints pattern in parallel across dependency types in English and Italian (Sprouse et al., 2016). For robustness of island effects and their independence from individual differences in working memory, see Sprouse et al. (2012).

Second, *super-additive interactions*: factors benign in isolation (NP-internal extraction; intervening nominals) combine to yield super-additive penalties across templates, tracking system-level dependency load rather than construction-specific features. Third, *operation invariance*: applying the same operation (e.g., *wh*-fronting) to diverse base clauses yields parallel licensing outcomes that generalize over compositional-structural principles, not template inventories.

This multi-level analysis has precedent in adjacent domains. Ekström et al. (2026) demonstrate that phonological systems – not just individual phonemes – exhibit signatures of culturally maintained cognitive tools, with system-level properties (inventory size clustering, marked-segment scaling) maintained by the same mechanisms operating across the phoneme inventory. Miller (2021) establishes that individual words are HPC kinds; the vocabulary system itself exhibits system-level properties (collocational patterns, semantic field structure) maintained at a higher organizational level. Our argument extends this pattern: just as phonemes and words support system-level organization, individual constructions support grammaticality as a system-level HPC kind.

4.1 THE PROPERTY CLUSTER

The grammaticality cluster centers on three core properties that tend to co-occur and mutually reinforce each other: morphosyntactic form–meaning pairings, compatibility between those pairings and contextual interpretation, and community entrenchment. These properties define objective grammatical status – whether the community licenses a pairing (§2) – as distinct from the subjective feeling of ungrammaticality, which integrates grammaticality with processing and social factors.

At the core lies the systematic pairing of morphosyntactic forms with meanings.³ These pairings

¹See Miller (forthcoming) for a full discussion of the ontological status of particular linguistic entities, including phrases and parts of speech.

²The stronger claim that grammaticality exhibits ontological EMERGENCE is explored in Nefdt and Reynolds (n.d.); for emergence in language more broadly, see Miller (2017). The present argument requires only the weaker mereological point.

³Reynolds (2025a) shows that even within such pairings, morphosyntactic and semantic properties can form distinct

aren't arbitrary but follow regular patterns: hierarchical structure maps to semantic composition, linear order maps to information structure, and morphological marking maps to grammatical features. When these pairings succeed, utterances feel grammatical. When they fail – as in **Furiously sleep ideas green colorless*, where no stable mapping emerges – grammaticality collapses. The systematicity of these pairings explains why speakers can productively extend patterns to novel items and why violations trigger consistent rejection across speakers.

Next, grammaticality requires alignment between what the morphosyntactic form *means* and what the utterance as a whole conveys. High compatibility occurs when tense morphology aligns with temporal reference, argument structure matches verb semantics, and agreement features show consistent values. The present perfect's incompatibility with past-time *yesterday* (**I've finished it yesterday*) exemplifies compatibility failure despite successful pairing – each element contributes its meaning, but those meanings clash. This dimension explains why structurally well-formed utterances can still be ungrammatical when meanings misalign.

Finally, conventionalization within a speech community determines which pairings count as established resources. Highly entrenched patterns show rapid processing, resist change, and transfer productively to new lexical items. The entrenchment difference explains why *sneaked* and *snuck* can coexist as grammatical variants while **bringed* remains impossible – the irregular *brought* has achieved complete entrenchment, blocking regularization. Importantly, entrenchment dissociates from raw frequency: high-exposure errors don't entrench, while some low-frequency patterns do when they instantiate licensed pairings.

These three properties cluster systematically while remaining partially independent. A construction can have intact pairings and high compatibility but lack entrenchment (the independent relative *whose*). Another can be highly entrenched with clear pairings but show compatibility failures (**I've finished it yesterday*). This partial independence while maintaining systematic co-occurrence is exactly what we expect from HPC kinds.

4.2 HOMEOSTATIC MECHANISMS MAINTAINING THE CLUSTER

The property cluster doesn't persist by accident. At least five identifiable causal mechanisms operating at distinct timescales actively maintain the clustering: bias-guided acquisition filters the hypothesis space during language learning, interactive alignment stabilizes patterns in real-time conversation, consolidation strengthens high-precision routines over days and years, prestige-weighted selection shapes which variants spread through communities, and iterated transmission filters out patterns that can't survive generational bottlenecks.

First, working memory constraints create a “now-or-never bottleneck” (Chater & Christiansen, 2010): fleeting linguistic input has to be processed and compressed immediately because unanalyzed signal can't be stored. Domain-general prediction machinery, modulated by attentional biases for hierarchical structure, extracts form–meaning pairings and rejects statistically unlicensed mappings during this incremental learning process. Children systematically avoid unattested patterns despite noisy input, revealing how the bottleneck forces early commitment to stable mappings that the community has conventionalized.

Second, in conversation, speakers converge automatically on syntactic frames and prune misalign-

HPC clusters maintained by different mechanisms. The present analysis focuses on the morphosyntactic side of form–meaning pairings as the foundation of grammaticality judgments.

ments through immediate repair (Pickering & Garrod, 2004). When violations occur, they trigger disfluencies and explicit corrections, continuously stabilizing community norms. This real-time mechanism explains why grammatical errors rarely propagate through conversations – the interactive system flags and repairs them before they can spread.

Third, over days and years, frequently accessed constructions become increasingly automated while rare patterns remain effortful. High-precision tokens are preferentially strengthened during memory consolidation, while low-precision ones decay. This produces heavy-tailed frequency distributions and explains why some constructions become so entrenched they resist change even when competing forms emerge.

Fourth, variants associated with high-status speakers diffuse faster through communities (Labov, 2001), pushing some constructions toward categorical acceptance while maintaining sociolinguistic stratification at gradient margins. This social mechanism explains why prescriptive norms can override frequency – stigmatizing common forms like double negatives – and why structural constraints often correlate with social evaluation.

Fifth, the transmission bottleneck across generations amplifies learnability biases and filters out mappings that exceed processing or memory constraints (Kirby et al., 2014). This multi-generational ratchet explains why structural prohibitions converge cross-linguistically even as surface patterns diverge. Only learnable, processable patterns survive iterated transmission, creating systematic gaps in the space of possible grammars.

Each of these five mechanisms selectively stabilizes part of the cluster: acquisition maintains pairings and compatibility, interaction enables online repair, and consolidation drives entrenchment, while prestige selection and iterated transmission ensure population-level stability. Figure 1 schematizes these relationships.

These mechanisms parallel those maintaining biological species, where developmental constraints, ecological pressures, and reproductive isolation jointly stabilize phenotypic clusters (Ereshefsky & Reydon, 2015). The key insight from an HPC conception of kinds (R. Boyd, 1991) is that such multi-level homeostasis can maintain kind-hood without requiring essences or sharp boundaries.

Cross-linguistic evidence confirms this homeostatic structure. Agreement violations elicit rapid P600 or LAN effects not only in Germanic and Romance languages but also in Hindi (Nevins et al., 2007), Basque (Zawiszewski & Friederici, 2009), and Arabic (Muralikrishnan & Idrissi, 2021). These convergent neural signatures show that the link between morphosyntax and processing travels with speakers across languages. Because acquisition, repair, and social transmission all funnel into maintaining reliable agreement constraints, the same violation types trigger comparable detection responses, supporting the claim that grammaticality is a projectable homeostatic kind.

4.3 PROJECTABILITY: PREDICTION AND EXPLANATION

A core aspect of an HPC account of kinds is that it allows for kinds that can support inductive and explanatory practices. Grammaticality passes three key tests: predicting judgment patterns, diachronic trajectories, and acquisition sequences.

First, the property cluster allows systematic prediction across construction types. Knowing someone rejects **Which car did you buy car?* (left-branch extraction) allows prediction they'll also reject **Which is charged phone?*, **What will they make kind?*, and **This is the person whose you met brother*. This projectability extends beyond surface similarity to abstract structural properties: all involve extraction from within a noun phrase. The cluster structure also explains why compatibility mismatches

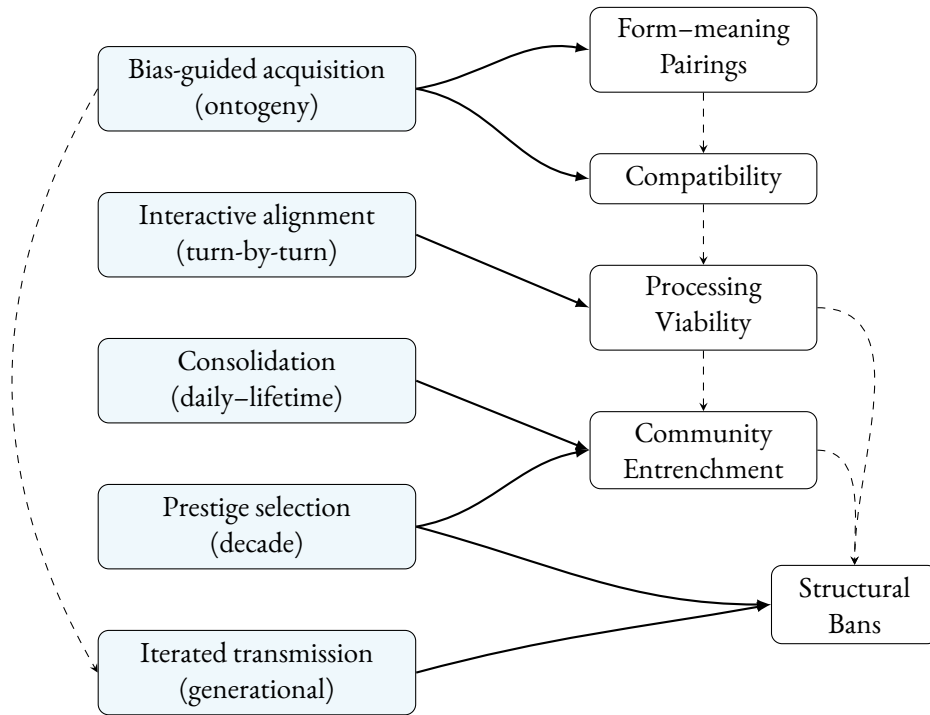


Figure 1: Homeostatic mechanisms stabilizing the grammaticality property cluster. Five mechanisms (left) operating at different timescales support a cluster of linguistic properties (middle), which collectively give rise to structural bans (right). Solid arrows denote primary causal support; dashed arrows show secondary modulation and feedback. Arrows to ‘Structural Bans’ indicate hypothesized boundary stabilization; they are compatible with bans that are partly autonomous. The diagram is heuristic, not a derivation. Processing Viability receives input from Interactive Alignment but also constrains what Acquisition and Iterated Transmission can maintain, creating a feedback loop between real-time comprehension and diachronic stability.

trigger rejection even when processing demands are low and the string is structurally well-formed. The cluster structure enables these inferences.

Second, grammaticalizing constructions follow predictable trajectories. When semantic transparency, social prestige, and analogical pressure favor a construction, entrenchment grows along characteristic S-shaped curves – slow initial adoption, rapid spread through the community, and eventual stabilization. Historical changes like *going to* → *gonna* exhibit exactly this pattern. The homeostatic mechanisms generate these trajectories: acquisition biases, interactive alignment, and prestige selection jointly determine growth rates, while consolidation and iterated transmission lock in successful innovations.

Third, second-language learners show predictable error trajectories. Initial reliance on L1 transfer gives way to overgeneralization of L2 patterns, followed by gradual approximation to target norms. These sequences reflect the homeostatic mechanisms at work: learners bring acquisition biases and processing constraints from L1, interactive feedback triggers repairs, and community entrenchment shapes which innovations persist. The systematicity of these trajectories supports grammaticality's status as a learnable kind with projectable properties.

In this way, HPC analysis explains why grammaticality supports robust generalization despite variation. The property cluster is stable enough to license prediction while remaining flexible enough to accommodate gradient judgments, community differences, and diachronic change. This combination – systematic projectability within bounded variation – is the hallmark of HPC kinds.

The analysis is falsifiable. Three patterns would undermine it. First, if structure-based predictions failed on new construction types while construction-specific labels succeeded, that would show grammaticality is just a list of templates rather than a system with cross-constructional properties. Second, if apparent cross-construction interactions turned out to be simple sums of independent construction-level factors, that would eliminate the part-whole claim. Third, if phonetic or discourse manipulations repaired classic structural bans as effectively as prosodic or structural interventions, that would collapse the regime boundaries. None of these patterns appear in the data, supporting the system-level HPC analysis.

5 BOUNDARY CASES AND CONTRAST CLASSES

A complete account needs to explain not just what falls within the kind but what falls outside it. Three contrast cases illuminate grammaticality's boundaries, demonstrating the importance of distinguishing objective grammatical status from subjective acceptability (§2). These cases also reveal how the same mechanisms instantiate differently under different constraint regimes.

5.1 PROCESSING OVERLOAD: PERFORMANCE FILTER, NOT KIND CRITERION

Center-embedded sentences like *The rat the cat the dog chased killed ate the cheese* challenge the grammaticality/acceptability distinction (Gibson, 2025, see esp. §§2.2.7–2.2.8). Each embedded clause has clear syntactic structure and semantic interpretation. The form–meaning mapping succeeds in principle – the construction is grammatical. But processing difficulty creates a strong feeling of ungrammaticality even when the licensing relation remains intact.

The difficulty is predictable from memory constraints, with each embedding adding measurable integration cost (Gibson, 2000). Processing difficulty can be reduced through prosody, context, or visual presentation, and speakers who can't process triple embedding still judge simpler versions grammatical, showing that their knowledge of the grammatical pattern remains intact. This dissociation

confirms our claim (§2) that processing modulates the subjective feeling of ungrammaticality rather than determining whether the community licenses the pairing.

When licensing and processing diverge – grammatical structure is licensed but computationally expensive – we observe consistent rejection in acceptability tasks while simpler instances reveal underlying competence. This dissociation is fully consistent with an HPC treatment. Processing constraints shape what patterns can be licensed (via acquisition filtering and transmission bottlenecks) while also filtering real-time accessibility independently. The HPC framework naturally accommodates this partial independence: the five mechanisms maintain the structural pattern as a licensed resource, while processing difficulty modulates how that pattern feels during comprehension without determining its licensing status. This supports treating grammaticality as an HPC kind with properties that can partially dissociate.

5.2 PHONOTACTICS: A PARALLEL HPC KIND

The bnick bnacks violates no grammatical constraints but sounds wrong to English speakers. The problem isn't morphosyntax but phonotactics: English doesn't allow word-initial /bn-/ clusters. Phonotactic well-formedness constitutes a distinct HPC kind from grammaticality, maintained by the same five mechanisms but operating under a different constraint regime.

Recent work establishes that phonological systems qualify as culturally maintained cognitive tools (Ekström et al., 2026). Phoneme inventories cluster predictably: most languages maintain 20 – 50 segments across unrelated families, despite the theoretical possibility of millions of combinations. Marked segments like /y/ (front rounded vowel) appear preferentially in larger vowel systems, rising from near-zero probability in 5-vowel systems to >0.75 in systems with 40+ vowels, while unmarked /i/ remains common throughout. These patterns survive genealogical controls and reflect identifiable stabilizing mechanisms: quantal regions that create articulatory attractors (Stevens, 1989), dispersion pressure that maintains distinctiveness (Liljencrants & Lindblom, 1972), perceptual magnets that bind variant productions to prototypes (Kuhl et al., 1992), and community norms that transmit systems across generations.

The five mechanisms we have identified – acquisition, interactive alignment, consolidation, prestige selection, and iterated transmission – operate identically to those maintaining grammaticality, but under a phonotactic constraint regime where articulatory geometry and auditory discriminability dominate. This produces different manifestations of each mechanism. Acquisition in phonotactics means learning articulatory targets and perceptual categories; in grammaticality it means learning dependency-respecting form-meaning mappings. Consolidation in phonotactics means motor entrenchment and gestural reduction; in grammaticality it means construction chunking under memory constraints. Prestige selection in phonotactics shapes accent and dialect norms; in grammaticality it enforces prescriptive rules about structural correctness. The mechanisms are the same; their instantiation differs because the constraint profiles differ.

This explains why certain factors matter dramatically in one regime but not another. Memory load and embedding depth constrain grammaticality but play minimal role in phonotactics. Vocal tract geometry and quantal stability dominate phonotactics but barely affect syntax (except through prosody). The distinction is empirical, not stipulated: interventions that ameliorate phonotactic oddness (careful articulation, slowed speech) don't license **Which did you buy car?*; interventions that improve grammatical acceptability (structural priming, prosodic chunking) don't fix phonotactic violations.

The phenomenology differs because different constraints are violated. **Which did you buy car?* triggers the sensation of ungrammaticality – something structurally wrong with the sentence. *The bnick bnacks* triggers phonotactic oddness – it sounds foreign or invented but not ungrammatical. The feeling difference reflects different constraint regimes being evaluated: compositional-interpretive constraints for grammaticality, articulatory-perceptual constraints for phonotactics.

Cross-linguistic evidence confirms the distinction. Languages converge on certain form-meaning constraints (no language allows completely free word order without morphological marking) but vary dramatically in phonotactic inventories. What counts as a well-formed syllable differs radically: Japanese restricts codas to nasals; English allows complex clusters; Hawaiian permits only open syllables. The phonotactic system remains stable within each language community; the patterns are maintained by the same five mechanisms but produce language-specific phonological kinds rather than universal grammatical constraints.

This reveals grammaticality's boundaries from one angle. Grammaticality licenses morphosyntactic structures and their compositional interpretation under memory and processing constraints. Phonotactics constrains sound-sequencing patterns under articulatory-perceptual constraints. Both are HPC kinds maintained by the same mechanisms. They're distinct because those mechanisms operate under different constraint regimes: compositional-interpretive constraints for grammaticality, articulatory-perceptual constraints for phonotactics.

5.3 LEXICAL SEMANTICS: FROM WORDS TO VOCABULARY SYSTEMS

Colorless green ideas sleep furiously is semantically anomalous but grammatical. Like phonotactics, lexical semantics constitutes a distinct HPC kind from grammaticality, maintained by the same five mechanisms but operating under yet another constraint regime. This distinction manifests at two levels: individual words and the vocabulary system as a whole.

Miller (2021) establishes that individual word-kinds – *table*, *Paris*, *egregious* – are HPC kinds. Each word maintains clustered properties: phonetic form, orthographic representation, semantic content, distributional patterns, grammatical category. Internal mechanisms (mental lexicon retrieval, production habits) and external mechanisms (community norms, editorial standards, frequency-based entrenchment) keep these properties bundled. Words can drift semantically – *egregious* shifted from positive (“distinguished”) to negative (“outrageous”) over centuries – while preserving enough co-variation for speakers to project from attested tokens to new ones. The word remains identifiable across the drift because stabilizing mechanisms maintain sufficient property clustering.

Ekström et al. (2026) demonstrates that phonological systems – not just individual phonemes – also constitute cognitive tools with system-level properties maintained by identifiable mechanisms. By parallel reasoning, the vocabulary system itself should qualify as a system-level HPC kind, distinct from but built upon individual word-kinds. The vocabulary system exhibits its own clustered properties: conventionalized collocations (*heavy rain*, not **strong rain* in English), structured semantic fields (kinship terms, color vocabularies), metaphor patterns (*ideas as buildings*, *time as money*), and selectional restrictions (*frighten* selects animate subjects; *elapse* requires temporal ones).

These system-level patterns are maintained by the same five mechanisms, but under a lexical-semantic constraint regime where conceptual compatibility and categorical structure dominate. This produces distinct manifestations. Acquisition in lexical semantics means learning concept-word alignments and collocational patterns; in grammaticality it means learning structural dependencies. Consolidation in lexical semantics means collocational entrenchment and metaphor conventionalization;

in grammaticality it means construction chunking. Prestige selection in lexical semantics shapes register and genre norms (latinate vocabulary in academic writing); in grammaticality it enforces prescriptive structural rules.

The crucial boundary emerges in *Colorless green ideas sleep furiously*. Each morphosyntactic element contributes its expected meaning: plural *-s* means ‘not one’, adjectival modification follows standard compositional rules, the verb-adverb combination instantiates a licensed pattern. The form-meaning pairing succeeds – grammaticality is satisfied. The oddity arises from conceptual incoherence: properties that contradict (*colorless* + *green*), subjects that violate selectional requirements (*ideas* + *sleep*), manners that don’t apply to the activity (*furiously* + *sleep*). Lexical-semantic constraints fail while grammatical operations succeed.

Lexical-semantic anomalies don’t block grammatical operations. We can form questions (*Do colorless green ideas sleep furiously?*), embed the clause (*John thinks that colorless green ideas sleep furiously*), and apply all standard syntactic transformations. The grammatical system remains intact. What fails is conceptual coherence – vocabulary-level constraints on which combinations make sense.

The phenomenological difference confirms the distinction. Grammatical violations like **Which did you buy car?* trigger the feeling of ungrammaticality – structural wrongness. Lexical-semantic anomalies trigger semantic oddness or conceptual incongruity. Speakers recognize the sentence as well-formed but conceptually bizarre. The difference reflects different constraint regimes: compositional mapping constraints (grammaticality) versus conceptual compatibility constraints (lexical semantics).

Cross-linguistic evidence reinforces the boundary. Languages converge on certain grammatical constraints (scope-bearing elements show restricted distributions universally) but vary dramatically in which conceptual combinations they conventionalize. *Ideas sleep* could work metaphorically in contexts where dormancy is culturally salient for abstract concepts. English conventionalizes *heavy rain*; other languages use ‘strong’ or ‘big’. The grammatical patterns remain relatively stable; the lexical-semantic acceptability varies with cultural conceptual structures.

5.3.1 Porous boundaries as an HPC prediction.

Because the same mechanisms operate under different constraint regimes, judgments sometimes integrate signals from multiple regimes simultaneously. This yields “leaky” phenomenology at interface cases. The choice between *a* and *an* is phonologically conditioned (vowel onset triggers *an*) yet grammatically obligatory (English requires determiners). French *le/la* embodies lexically arbitrary gender under grammatical compulsion. Collocational preferences (*heavy rain* versus **strong rain*) ride on licensed argument-structure frames, blending lexical entrenchment with grammatical well-formedness.

On our view, such porosity is not an embarrassment but the expected footprint of overlapping constraint regimes. The mechanisms don’t just operate in parallel – they genuinely interact, with constraints from one regime shaping realizations in another.

Consider *a/an* alternation. The choice is phonologically conditioned: /ən/ appears before vowel-initial words, avoiding the dispreferred /ə/ + vowel sequence. This looks like a phonotactic pressure – articulatory economy drives the alternation. But the alternation occurs within a grammatically obligatory frame: English requires determiners with singular count nouns. And unlike typical phonological processes (which are often optional or variable), *a/an* selection is categorical and obligatory, patterned like agreement morphology. The result is a hybrid: phonotactic constraints shape

the morphological realization of a grammatical requirement. Acquisition extracts both the determiner requirement and its allomorphic conditioning; prestige mechanisms enforce both as obligatory; consolidation entrenches the pattern. The feeling of wrongness in **a apple* integrates phonotactic awkwardness (the /ə/-vowel sequence) with morphological error (wrong allomorph selection).

Collocation shows a different interaction architecture. English licenses adjective-noun modification productively – the grammatical frame permits **heavy rain**, **strong rain**, **intense rain**. But community entrenchment selects among these possibilities: *heavy rain* has become conventional while **strong rain** remains marginal despite being grammatical and conceptually coherent. The mechanisms work jointly. Acquisition extracts both the productive modification frame and the collocational preferences within it. Consolidation preferentially strengthens high-frequency pairings like *heavy rain*, making them automatic while alternatives remain effortful. Interactive alignment propagates the preference: speakers converge on *heavy* because that’s what the community says. Prestige selection can reinforce particular collocations in different registers. The key point is that grammatical licensing provides the FRAME within which lexical entrenchment operates: you can’t conventionalize a collocation the grammar doesn’t license. Conversely, high-frequency collocations reinforce the grammatical frames that host them – the adjective-noun modification construction is strengthened every time someone says *heavy rain*.

The property cluster for *heavy rain* includes: grammatical licensing of the modification frame, high token frequency of the specific pairing, conceptual coherence (heaviness maps to precipitation intensity), and community conventionalization. These properties co-occur because the mechanisms maintain them jointly. The feeling with **strong rain* is ambiguous: grammatically licensed (so not “wrong”), conceptually coherent (so not “anomalous”), but unconventional (so not “natural”). This ambiguity reflects genuine regime overlap – the grammatical system licenses the pattern while the lexical-semantic system hasn’t entrenched it.

The HPC framework predicts this porosity. Constraint regimes are not hermetically sealed because the same mechanisms operate across all regimes simultaneously. When phonotactic, lexical-semantic, and grammatical constraints align, patterns entrench strongly. When they interact – where one regime provides a frame and another selects within it, or where gradual strengthening shifts a pattern from one regime toward another – the resulting properties show hybrid clustering. This porosity reflects the underlying causal architecture rather than undermining the kind distinctions.

6 CHALLENGES FROM REDUCTION AND FRAGMENTATION

Two challenges to the HPC analysis deserve response: the reductionist claim that grammaticality reduces to frequency or processing, and the pluralist claim that it splinters into multiple distinct kinds.

Reduction to frequency or processing. Reductionists hold that grammaticality reduces to usage frequency, processing ease, or some combination of the two. However, neither alone suffices. Rare constructions can be grammatical (*This is Joan, [a friend of whose] you met*),⁴ and frequent L2 errors almost never become grammatical through repetition (Han, 2004). Frequency contributes via entrenchment once licensing is in place, but does not determine it. Conversely, multiply center-embedded sentences are unprocessable but grammatical, while simple violations like **I have two cat* process easily but remain ungrammatical. The HPC framework incorporates frequency (through

⁴see Reynolds (2025b)

entrenchment) and processing (through mechanisms that shape licensing and affect feeling) while maintaining grammaticality as a genuine kind.

Fragmentation into multiple kinds. Pluralists argue that “grammaticality” names multiple distinct phenomena – morphological, syntactic, and semantic well-formedness – that don’t belong together. While the internal complexity is real, these phenomena show sufficient unity. Agreement features are morphological but have syntactic distribution and semantic interpretation. Children acquire these supposedly distinct systems in integrated fashion, and the same learning, processing, and social mechanisms maintain all aspects of the cluster. The HPC framework accommodates internal complexity while maintaining kind unity, just as *mammal* includes monotremes and marsupials within a single kind.

7 IMPLICATIONS

Treating grammaticality as an HPC kind has implications for linguistic methodology, theory, and typology:

7.1 METHODOLOGY: JUDGMENT TASKS, CORPORA, ERP

The HPC analysis validates methodological pluralism while explaining why different methods sometimes diverge. Acceptability judgments tap the subjective feeling, which includes both grammatical and processing factors. High inter-speaker agreement on clear cases validates the method while variation on boundary cases is expected. The framework predicts when convergence will be high (clear violations of morphosyntactic pairing) and when divergence will appear (processing difficulties, novel constructions undergoing entrenchment).

Corpus frequencies reflect production biases including processing cost, social indexicality, and discourse function alongside grammaticality. Absence from corpora doesn’t entail ungrammaticality – rare but licensed patterns may simply lack communicative motivation. Conversely, high corpus frequency doesn’t guarantee licensing, as demonstrated by persistent high-exposure errors that never entrench (Wagers et al., 2009).

Neural measures like ERP validate the cognitive reality of constraint-violation detection. P600 effects for violations appear cross-linguistically (Muralikrishnan & Idrissi, 2021; Nevins et al., 2007; Zawiszewski & Friederici, 2009), showing grammaticality’s psychological reality while demonstrating it’s not reducible to any single neural signature. The framework predicts when methods will converge (clear morphosyntactic violations) and when they’ll diverge (processing-heavy but licensed structures, socially stratified variants).

7.2 THEORETICAL: RECONCILING GENERATIVE AND USAGE-BASED EVIDENCE

The HPC framework suggests generative and usage-based approaches capture different aspects of the same kind rather than competing theories of distinct phenomena.

The generative research program has identified robust patterns – locality effects, island constraints, structure-sensitive dependencies – that any adequate theory must explain. These patterns are real: they show up consistently in acceptability judgments, resist frequency-based erosion, and recur cross-linguistically. The usage-based tradition has identified equally robust patterns – frequency effects, gradient acceptability, S-curve diffusion – that also demand explanation. Both sets of patterns belong in the cluster.

Where the approaches diverge is on *explanatory mechanisms*. Generative accounts appeal to innate principles or autonomous syntactic constraints. Usage-based accounts trace patterns to learning biases, processing limitations, and communicative pressures. The HPC framework requires causal mechanisms, and the five we identify – acquisition under a now-or-never bottleneck, interactive alignment, consolidation, prestige selection, iterated transmission – are domain-general cognitive and social processes. These mechanisms explain both categorical patterns (sharp boundaries emerge where learning bottlenecks force early commitments) and gradient patterns (acceptability continua emerge where consolidation and prestige operate).

7.3 TYPOLOGY: PREDICTING WHICH “EXOTIC” GRAMMARS CAN EXIST

The HPC analysis makes testable predictions about possible human languages by identifying which variation the homeostatic mechanisms can accommodate and which they cannot.

Languages can vary in which semantic distinctions receive morphosyntactic marking (tense, aspect, evidentiality, noun class), how social factors influence entrenchment (prestige-driven standardization versus egalitarian variation), and where processing limitations create acceptability gradients (embedding depth tolerance, dependency locality effects). This variation reflects different equilibria the same mechanisms can maintain under different social ecologies and communicative pressures.

Given current learnability and processing mechanisms, languages effectively cannot lack systematic form-meaning pairings entirely, have grammars unlearnable through the standard acquisition mechanism, or violate processing constraints beyond recoverable limits. These prohibitions follow from grammaticality’s homeostatic mechanisms. Any pattern incompatible with child acquisition (violating the now-or-never bottleneck), adult processing (exceeding working memory capacity), or social transmission (requiring perfect fidelity across generations) cannot stabilize as a grammatical system.

The framework thus predicts a bounded space of possible grammars – neither unlimited typological diversity nor rigid universal constraints, but systematic variation within mechanistically determined boundaries.

8 CONCLUSION

This paper has argued that grammaticality is a genuine kind, realized as a homeostatic property-cluster (HPC) kind. Like biological species, grammaticality exhibits clustered properties maintained by identifiable mechanisms without reducing to essential features. The analysis resolves longstanding puzzles about gradience, variation, and the competence/performance distinction while validating linguistics’ methodological practices.

The HPC framework reveals grammaticality as neither purely natural nor purely social but as emerging from the interaction of cognitive constraints, social conventions, and functional pressures. This perspective suggests that debates between formalist and functionalist approaches may rest on false dichotomies – grammaticality is the kind of complex phenomenon that requires both types of explanation.

Grammaticality plays a dual role in this framework. As a category, it is the HPC kind we have described: the maintained clustering of form–meaning pairings, compatibility relations, and community entrenchment. As a mechanism, it is the enforcement system for other linguistic kinds. The category NOUN is maintained partly because nounhood is grammatical – because using a word in a

noun slot with noun morphology triggers no mismatch signal. Grammaticality stabilizes the form–meaning couplings that constitute all other categories. Like the immune system, which is both a biological category and a mechanism that maintains other systems, grammaticality is both maintained and a maintainer. This dual role is characteristic of homeostatic systems, not paradoxical.

By taking grammaticality’s ontology seriously, we gain a framework that explains why human languages exhibit the particular mix of universality and diversity they do. The homeostatic mechanisms that maintain grammaticality also constrain its variation: only patterns compatible with acquisition, processing, and social transmission can stabilize. What resists at a given moment is entrenched convention, not timeless prohibition – but convention maintained by mechanisms powerful enough to make the distinction matter.

REFERENCES

- Bickel, B. (2007). Typology in the 21st century: Major current developments. *Linguistic Typology*, 11(1), 239–251. <https://doi.org/10.1515/LINGTY.2007.018>
- Bock, J. K. (1986). Syntactic persistence in language production. *Cognitive Psychology*, 18(3), 355–387. [https://doi.org/10.1016/0010-0285\(86\)90004-6](https://doi.org/10.1016/0010-0285(86)90004-6)
- Boyd, R. (1991). Realism, anti-foundationalism and the enthusiasm for natural kinds. *Philosophical Studies*, 61, 127–148. <https://doi.org/10.1007/BF00385837>
- Boyd, R. N. (1990). Realism, approximate truth, and philosophical method. In C. W. Savage (Ed.), *Scientific theories* (pp. 355–391, Vol. 14). University of Minnesota Press.
- Boyd, R. N. (1999a). Homeostasis, species, and higher taxa. In R. A. Wilson (Ed.), *Species: New interdisciplinary essays* (pp. 141–185). MIT Press.
- Boyd, R. N. (1999b). Kinds, complexity and multiple realization. *Philosophical Studies*, 95, 67–98. <https://doi.org/10.1023/A:1004511407133>
- Bybee, J. (2006). From usage to grammar: The mind's response to repetition. *Language*, 82(4), 711–733. <https://doi.org/10.1353/lan.2006.0186>
- Chater, N., & Christiansen, M. H. (2010). Language acquisition meets language evolution. *Cognitive Science*, 34(7), 1131–1157. <https://doi.org/10.1111/j.1551-6709.2009.01049.x>
- Croft, W. (2001). *Radical construction grammar: Syntactic theory in typological perspective*. Oxford University Press. <https://doi.org/10.1093/acprof:oso/9780198299554.001.0001>
- Eckert, P. (2012). Three waves of variation study: The emergence of meaning in the study of sociolinguistic variation. *Annual Review of Anthropology*, 41, 87–100. <https://doi.org/10.1146/annurev-anthro-092611-145828>
- Ekström, A. G., Tennie, C., Moran, S., & Everett, C. (2026). The phoneme as a cognitive tool. *Topics in Cognitive Science*, 18(1), 43–61. <https://doi.org/10.1111/tops.70021>
- Ereshefsky, M., & Reydon, T. A. C. (2015). Scientific kinds. *Philosophical Studies*, 172(4), 969–986. <https://doi.org/10.1007/s11098-014-0301-4>
- Fanselow, G. (2021). Acceptability, grammar, and processing. In G. Goodall (Ed.), *The cambridge handbook of experimental syntax* (pp. 118–153). Cambridge University Press. <https://doi.org/10.1017/9781108569620.006>
- Francis, E. J. (2022). *Gradient acceptability and linguistic theory*. Oxford University Press. <https://doi.org/10.1093/oso/9780192898944.001.0001>
- Gibson, E. (1998). Linguistic complexity: Locality of syntactic dependencies. *Cognition*, 68(1), 1–76. [https://doi.org/10.1016/S0010-0277\(98\)00034-1](https://doi.org/10.1016/S0010-0277(98)00034-1)
- Gibson, E. (2000). The dependency locality theory: A distance-based theory of linguistic complexity. In A. Marantz, Y. Miyashita & W. O'Neil (Eds.), *Image, language, brain* (pp. 95–126). MIT Press. <https://doi.org/10.7551/mitpress/3654.003.0008>
- Gibson, E. (2025). *Syntax: A cognitive approach*. MIT Press.
- Goldberg, A. E. (1995). *Constructions: A Construction Grammar approach to argument structure*. University of Chicago Press.
- Han, Z. (2004). *Fossilization in adult second language acquisition*. Multilingual Matters. <https://doi.org/10.21832/9781853596889>
- Hawkins, J. A. (2004). *Efficiency and complexity in grammars*. Oxford University Press. <https://doi.org/10.1093/acprof:oso/9780199252695.001.0001>

- Heim, I., & Kratzer, A. (1998). *Semantics in generative grammar*. Blackwell.
- Kaan, E., Harris, A., Gibson, E., & Holcomb, P. (2000). The P600 as an index of syntactic integration difficulty. *Language and Cognitive Processes*, 15(2), 159–201. <https://doi.org/10.1080/016909600386084>
- Khalidi, M. A. (2013). *Natural categories and human kinds: Classification in the natural and social sciences*. Cambridge University Press. <https://doi.org/10.1017/CBO9780511998553>
- Kirby, S., Griffiths, T. L., & Smith, K. (2014). Iterated learning and the evolution of language. *Current Opinion in Neurobiology*, 28, 108–114. <https://doi.org/10.1016/j.conb.2014.07.014>
- Klein, W. (1992). The present perfect puzzle. *Language*, 68(3), 525–552. <https://doi.org/10.2307/415793>
- Kratzer, A. (1998). More structural analogies between pronouns and tenses. In D. Strolovitch & A. Lawson (Eds.), *Proceedings of SALT 8* (pp. 92–110). <https://doi.org/10.3765/salt.v8i0.2808>
- Kripke, S. A. (1980). *Naming and necessity*. Harvard University Press.
- Kuhl, P. K., Williams, K. A., Lacerda, F., Stevens, K. N., & Lindblom, B. (1992). Linguistic experience alters phonetic perception in infants by 6 months of age. *Science*, 255(5044), 606–608. <https://doi.org/10.1126/science.1736364>
- Labov, W. (2001). *Principles of linguistic change, Volume 2: Social factors*. Wiley-Blackwell. <https://doi.org/10.1002/9781444327496>
- Liljencrants, J., & Lindblom, B. (1972). Numerical simulation of vowel quality systems: The role of perceptual contrast. *Language*, 48(4), 839–862. <https://doi.org/10.2307/411991>
- Lu, J., Frank, M., & Degen, J. (2024). A meta-analysis of syntactic satiation in extraction from islands. *Glossa Psycholinguistics*, 3(1), 1–33. <https://doi.org/10.5070/g60111425>
- Mason, R., & Ritchie, K. (2020). Social ontology. In R. Bliss & J. Miller (Eds.), *The routledge handbook of metametaphysics*. Routledge. <https://doi.org/10.4324/9781315112596-24>
- Miller, J. T. M. (2017). Language and ontological emergence. *Philosophica*, 91(1), 105–143. <https://doi.org/10.21825/philosophica.82118>
- Miller, J. T. M. (2021). Words, species, and kinds. *Metaphysics*, 4(1), 18–31. <https://doi.org/10.5334/met.70>
- Miller, J. T. M. (forthcoming). *Words and other linguistic entities*. Oxford University Press.
- Muralikrishnan, R., & Idrissi, A. (2021). Salience-weighted agreement feature hierarchy modulates language comprehension. *Cortex*, 141, 168–189. <https://doi.org/10.1016/j.cortex.2021.03.029>
- Nefdt, R., & Reynolds, B. (n.d.). *Grammar and emergence* [in preparation].
- Nevins, A., Dillon, B., Malhotra, S., & Phillips, C. (2007). The role of feature-number and feature-type in processing Hindi verb agreement violations. *Brain Research*, 1164, 81–94. <https://doi.org/10.1016/j.brainres.2007.05.058>
- Newmeyer, F. J. (2005). *Possible and probable languages: A generative perspective on linguistic typology*. Oxford University Press. <https://doi.org/10.1093/acprof:oso/9780199274338.001.0001>
- Osterhout, L., & Holcomb, P. J. (1992). Event-related brain potentials elicited by syntactic anomaly. *Journal of Memory and Language*, 31(6), 785–806. [https://doi.org/10.1016/0749-596X\(92\)90039-Z](https://doi.org/10.1016/0749-596X(92)90039-Z)
- Phillips, C., Wagers, M. W., & Lau, E. F. (2011). Grammatical illusions and selective fallibility in real-time language comprehension. In J. T. Runner (Ed.), *Experiments at the interfaces* (pp. 147–180, Vol. 37). Brill. https://doi.org/10.1163/9781780523750_006

- Pickering, M. J., & Garrod, S. (2004). Toward a mechanistic psychology of dialogue. *Behavioral and Brain Sciences*, 27(2), 169–190. <https://doi.org/10.1017/S0140525X04000056>
- Putnam, H. (1975). The meaning of “meaning”. In K. Gunderson (Ed.), *Language, mind, and knowledge* (pp. 131–193, Vol. 7). University of Minnesota Press.
- Reynolds, B. (2025a). *Definiteness and deitality in English: A homeostatic property cluster account* [Manuscript, LingBuzz/009369]. <https://ling.auf.net/lingbuzz/009369>
- Reynolds, B. (2025b). *Independent relative whose: From syntactic gap to pragmatic constraint* [In preparation].
- Schmid, H.-J. (2020). *The dynamics of the linguistic system: Usage, conventionalization, and entrenchment*. Oxford University Press. <https://doi.org/10.1093/oso/9780198814771.001.0001>
- Schütze, C. T. (2016). *The empirical base of linguistics: Grammaticality judgments and linguistic methodology*. Language Science Press. <https://doi.org/10.17169/langsci.b89.100>
- Snyder, W. (2000). An experimental investigation of syntactic satiation effects. *Linguistic Inquiry*, 31(3), 575–582. <https://doi.org/10.1162/002438900554479>
- Snyder, W. (2022). On the nature of syntactic satiation. *Languages*, 7(1), 38. <https://doi.org/10.3390/languages7010038>
- Sorace, A., & Keller, F. (2005). Gradience in linguistic data. *Lingua*, 115(11), 1497–1524. <https://doi.org/10.1016/j.lingua.2004.07.002>
- Sprouse, J., Caponigro, I., Greco, C., & Cecchetto, C. (2016). Experimental syntax and the variation of island effects in English and Italian. *Natural Language & Linguistic Theory*, 34(1), 307–344. <https://doi.org/10.1007/s11049-015-9286-8>
- Sprouse, J., Wagers, M., & Phillips, C. (2012). A test of the relation between working-memory capacity and syntactic island effects. *Language*, 88(1), 82–123. <https://doi.org/10.1353/lan.2012.0004>
- Stevens, K. N. (1989). On the quantal nature of speech. *Journal of Phonetics*, 17(1–2), 3–45. [https://doi.org/10.1016/S0095-4470\(19\)31520-7](https://doi.org/10.1016/S0095-4470(19)31520-7)
- Tomasello, M. (2003). *Constructing a language: A usage-based theory of language acquisition*. Harvard University Press.
- Wagers, M. W., Lau, E. F., & Phillips, C. (2009). Agreement attraction in comprehension: Representations and processes. *Journal of Memory and Language*, 61(2), 206–237. <https://doi.org/10.1016/j.jml.2009.04.002>
- Zawiszewski, A., & Friederici, A. D. (2009). Processing canonical and non-canonical sentences in Basque: The case of object–verb agreement as revealed by event-related brain potentials. *Brain Research*, 1284, 161–179. <https://doi.org/10.1016/j.brainres.2009.05.099>